

# Topics

## Frontend

| CATEGORY         | CHAPTERS                                                                    | OVERVIEW                                                                                                                                                                                                                                                                      |
|------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Introduction     | What is Web?<br>Client-Server Architecture<br>Setting Developer Environment | Understanding how and who built the web<br><br>General architecture used by websites; requires & response<br><br>Setting our environment on our laptop/computer<br>where we'll do coding                                                                                      |
| HTML             | Structure<br>Tags in HTML<br>Block v/s Inline<br>Tables<br>Forms            | How to create the structure of a web page<br><br>Learning about various tags in HTML like <h1>, <p>, <a>, <img> etc<br><br>Understand the difference between inline and block HTML<br>elements<br><br>Learn to create tables in HTML<br><br>Learn about forms and form fields |
| Intro to CSS     | Introduction                                                                | What is CSS & how to use it in HTML, different styles of writing                                                                                                                                                                                                              |
| Selectors in CSS | Understanding Selectors<br>Selector Specificity                             | Element, Class & Id selectors etc., combinators, pseudo<br>classes, pseudo elements, specificity in CSS<br><br>Understanding the specificity & priority of CSS selectors                                                                                                      |

| CATEGORY           | CHAPTERS             | OVERVIEW                                                                                |
|--------------------|----------------------|-----------------------------------------------------------------------------------------|
| Styling with CSS   | Box Model            | Understanding the CSS box model                                                         |
|                    | CSS Units            | Learning about various CSS units used to style HTML elements, absolute & relative       |
| More CSS           | CSS Transition       | Understanding element transitions in CSS along with shorthand                           |
|                    | CSS Transforms       | Understanding element transformations in CSS along with shorthand                       |
| Flexbox            | Intro to Flex        | Understanding flexbox layout, cross axis, main axis etc.                                |
|                    | Flex properties      | flexbox direction, justify content, align items, align self, flexwrap, flex sizing,etc. |
| Responsive Designs | Media Queries        | Learn about Media Queries & Viewport                                                    |
| Bootstrap          | Frontend frameworks  | What are frontend frameworks and how to use one                                         |
|                    | Components           | Using various bootstrap elements like Navbar, buttons, cards etc                        |
|                    | Layouts(Grid system) | Learning about grid system of bootstrap                                                 |
| Tailwind CSS       | What is Tailwind?    | Understanding Tailwind as a Framework                                                   |

| CATEGORY                   | CHAPTERS                                                                           | OVERVIEW                                                                                                                                                                       |
|----------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tailwind CSS               | Components<br>Creating Responsive Designs                                          | Covering button, navbar, fonts, margin, padding etc.<br>Understanding responsiveness in tailwind, @apply, @layer etc.                                                          |
| Major Project              | CSS Major Project                                                                  | Focus on using concepts we have learn to build our project                                                                                                                     |
| Starting with Javascript   | Intro to JS<br>Variables, operators, conditional, loops                            | What is JS and use of JS<br>Learning the basics of the language                                                                                                                |
| Functions and Arrays in JS | Scope<br>Functions expressions v/s<br>Function declaration<br>Arrays and its usage | Understanding scope in JS<br>Difference between function expression and declaration<br>What are arrays and using array functions like splice, slice etc.                       |
| Objects and Timing Events  | Intro to Objects<br>Object functions                                               | What are objects, how to create them and using different notations to access object's data<br>Learn to iterate over objects, delete object properties, creating nested objects |
| Understanding DOM          | DOM                                                                                | Understanding DOM, what it is, how to access elements from the DOM                                                                                                             |

| CATEGORY                    | CHAPTERS           | OVERVIEW                                                                        |
|-----------------------------|--------------------|---------------------------------------------------------------------------------|
| Understanding DOM           | Events             | How to manipulate DOM events in JS                                              |
| Closures                    | IIFE               | What are immediately invoked function expressions                               |
|                             | Closures           | What are closures and its application                                           |
|                             | Arrow functions    | Learning about arrow functions and bindings in arrow functions                  |
| Constructors and Prototypes | “this” keyword     | How does the “this” keyword works in JS                                         |
|                             | Prototypes         | Discussing what are prototypes in JS, why do we use them and its application    |
|                             | Class              | Learning about using classes in JS and how to deal with class inheritance in JS |
| Ashynchronous JavaScript    | Promises, Callback | What are promises and callbacks in Javascript, Why to use                       |
|                             | Timed Events       | What is setTimeout, Event loops in javascript                                   |
|                             | Async Await        | What are Async Await in Javascript, Why that is important                       |
| Ajax                        | Intro to AJAX      | What are async requests, what is API and JSON                                   |

| CATEGORY      | CHAPTERS           | OVERVIEW                                                                          |
|---------------|--------------------|-----------------------------------------------------------------------------------|
| Promises      | Handling promises  | What is a promise, how do we use promises and chaining promises                   |
| Git           | Intro to Git       | What is git and why it's helpful                                                  |
|               | Branches           | Exploring branches in Git. How to create branches.                                |
|               | Git workflow       | Understanding push, commits, pull requests and using git for teams and individual |
| Terminal      | Mastering Terminal | Directories, Commands, paths, operations on files etc.                            |
| Major Project | JS Major Project   | Create something classic by using the concept learn in JS                         |



| CATEGORY                 | CHAPTERS               | OVERVIEW                                                              |
|--------------------------|------------------------|-----------------------------------------------------------------------|
| Node.js                  | Intro to Node          | Introduction to the course, hello world with nodejs                   |
|                          | Setting up             | Setting up tools and the project                                      |
| Writing Our First Server | Intro to servers       | What are servers and how one can use them                             |
|                          | Setting up node server | Beginning the project by setting up the very first node server        |
|                          | nodemon                | Introducing nodemon to monitor changes made to the server             |
| Creating Express Apps    | MVC                    | MVC architecture for our server                                       |
|                          | Express                | What are frameworks, using express with node                          |
|                          | Ejs                    | What are template engines, setting up and working with Ejs            |
|                          | Middleware             | What is a middleware and how to use one                               |
| Intro to Databases       | Database               | What are databases & why do we need them                              |
|                          | SQL                    | what is SQL, SQL queries etc.                                         |
| MongoDb                  | MongoDB                | What is MongoDB, how to use it and setting up MongoDB for the project |
|                          | DB operations          | CRUD operations for MongoDB                                           |

## Backend

| CATEGORY     | CHAPTERS                    | OVERVIEW                                                                                                        |
|--------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| MongoDb      | Mongoose                    | Linking MongoDB using Mongoose                                                                                  |
| Mega Project | Working on our Mega Project | APIs, error handling, validation, express router, authentication, deployment & many more concepts to be covered |

## React

| CATEGORY      | CHAPTERS                   | OVERVIEW                                                                                                       |
|---------------|----------------------------|----------------------------------------------------------------------------------------------------------------|
| React         | Components, Styling & more | What is React, installation, react components, styling in react, component lifecycle methods, Material UI etc. |
| React Project | React based Project        | Using the concepts we have learnt to build our project                                                         |